Trust Chain: Blockchain Powered E-Commerce Reputation System

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***Abstract*— Trust and reputation are critical components in modern e-commerce platforms, where users often rely on feedback and ratings to make informed decisions. However, traditional reputation systems are prone to manipulation, identity fraud, and a lack of transparency due to their centralized architecture. To address these issues, this paper proposes Trust Chain, a blockchain-powered e-commerce reputation system that ensures secure, tamper-proof, and privacy-preserving feedback management. Leveraging Hyperledger Indy for decentralized identity and Hyperledger Fabric for permissioned blockchain infrastructure, the system enables users to submit verifiable reviews without revealing personal information. It incorporates Verifiable Credentials, Zero-Knowledge Proofs (ZKP), and smart contracts to authenticate feedback, automate reputation scoring, and prevent fraudulent activities. Additionally, a token-based incentive mechanism is introduced to encourage honest participation. The proposed solution enhances trust and transparency in e-commerce by providing a decentralized and scalable reputation framework suitable for real-world deployment.**

***Keywords*— Hyperledger Indy, Hyperledger Fabric, Blockchain**

I. INTRODUCTION

This The rapid growth of e-commerce has transformed the global marketplace, enabling consumers and merchants to interact across geographical boundaries with unprecedented ease. A cornerstone of this digital ecosystem is the reputation system, which helps establish trust between anonymous parties through user-generated feedback and ratings. However, existing centralized reputation mechanisms are prone to manipulation, censorship, and lack of transparency. Fake reviews, biased moderation, and identity fraud undermine the reliability of such systems, posing significant risks to users and businesses alike.

Blockchain technology, with its inherent properties of decentralization, immutability, and transparency, offers a promising solution to these challenges. By leveraging blockchain, reputation data can be securely stored and verified without reliance on a single controlling entity. This paper proposes "Trust Chain," a blockchain-powered e-commerce reputation framework that integrates Hyperledger Indy for decentralized identity management and Hyperledger Fabric for secure feedback handling. The system ensures privacy-preserving, tamper-proof, and transparent feedback processing, addressing the shortcomings of conventional reputation models. Through this approach, the proposed system aims to establish a scalable and trustworthy infrastructure for enhancing user confidence and integrity in e-commerce transactions.

II. RELATED WORK

In [1], Sun et al. proposed a blockchain-based feedback token system that allows only verified buyers to submit reviews, mitigating ballot-stuffing attacks. However, the solution was limited by high transaction fees on public blockchains. In a similar direction,

Li et al. [2] developed RepChain, a privacy-preserving reputation model that employs blind signatures to anonymize users. While effective in protecting identities, the model lacked incentive mechanisms for honest feedback.

The DTrust framework by Dhakal et al. [3] introduced a decentralized trust management system using Ethereum

smart contracts. It automated review validation but was constrained by the cost of on-chain data processing and limited scalability. EthReview by Zulfiqar et al. [4] enhanced this approach by integrating smart contracts for secure product reviews. However, it failed to ensure the privacy of reviewers.
Kugblenu and Vuorimaa [5] explored the use of permissioned blockchains for e-commerce reputation management. Their model improved performance and governance but lacked strong cryptographic verification of feedback authenticity. Another relevant work is Beaver [6], a decentralized e-commerce platform that includes a built-in reputation system. Despite its full decentralization, it requires an entirely new marketplace infrastructure, limiting its adaptability to existing platforms.
The proposed system, Trust Chain, builds upon these foundations by integrating Hyperledger Indy for decentralized identity and verifiable credentials, and Hyperledger Fabric for smart contract execution and tamper-proof review storage. It addresses the shortcomings of prior models by offering scalability, privacy, and incentive mechanisms within a permissioned blockchain architecture.

III. METHODLOGY

The proposed system, **Trust Chain**, combines permissioned blockchain architecture with cryptographic verification and smart contracts to build a **transparent and privacy-preserving e-commerce reputation system**. The system leverages **Hyperledger Fabric** for blockchain network implementation and **Hyperledger Indy** for decentralized identity management using **Verifiable Credentials (VCs)** and **Zero-Knowledge Proofs (ZKP)**.

A. Tools and Technologies Used

- **Hyperledger Fabric**: A modular, permissioned blockchain framework used for deploying smart contracts (chain code), managing transaction consensus, and maintaining an immutable ledger of buyer-seller interactions and feedback.
- **Hyperledger Indy**: A distributed ledger purpose-built for decentralized identity management, enabling the generation and verification of Verifiable Credentials.
- **Zero-Knowledge Proofs (ZKP)**: Used to validate credentials without disclosing user identity during the review process.
- **Smart Contracts (Chaincode)**: Developed using Go/JavaScript to automate reputation score calculation and validation logic on the blockchain.
- **Tokenization System**: Two types of tokens are utilized — **Feedback Tokens (FT)** for review submission and **Discount Tokens (DT)** as incentives for verified reviewers.

B. Identity and Credential Verification

1. Upon registration, the **Issuer (trusted authority)** generates a Verifiable Credential for each user (buyer/seller).
2. Users receive **Decentralized Identifiers (DIDs)** linked to their credentials.
3. During feedback submission, users generate a **ZKP presentation** using Hyperledger Indy to prove ownership of the credential without revealing private details.

C. Feedback Verification Algorithm

The feedback verification and reward process consist of the following steps:

1. Buyer makes a purchase and is issued a **Feedback Token (FT)**.
2. Buyer submits feedback along with a ZKP-protected credential.
3. Smart contract (chaincode) on Hyperledger Fabric:
   - Validates the FT.
   - Verifies ZKP authenticity.
   - Stores the feedback immutably in the blockchain.
4. If all checks pass, a **Discount Token (DT)** is generated and issued to the buyer.

D. Reputation Score Calculation Algorithm

The reputation score **R** for a seller is calculated as:

$$R = \sum_{i=1}^{n} (w_i \cdot f_i)$$

Where:

- fi: Feedback rating from buyer *i* (on a normalized scale)
- wi: Weight factor based on buyer's credibility (derived from Verifiable Credentials)
- n: Total number of verified feedback entries

F. System Security & Privacy

The system ensures:

- **Tamper-proof feedback** through immutable ledger storage.
- **Anonymity** via ZKP and decentralized identifiers.
- **Selective disclosure** of attributes through verifiable credentials.
- **Replay attack prevention** using one-time tokens and timestamped signatures.

IV. RESULTS

The proposed system, **Trust Chain**, was successfully implemented using **Hyperledger Indy** and **Hyperledger Fabric** to evaluate its effectiveness in securing and managing e-commerce reputation. The integration of verifiable credentials with Zero-Knowledge Proofs (ZKP) ensured that only legitimate, privacy-preserving reviews were accepted. Feedback and reputation data were stored immutably on the blockchain, ensuring tamper-proof transparency.

Performance testing demonstrated the system's ability to:

- **Authenticate users** securely via decentralized identity credentials.
- **Prevent fake reviews** using feedback tokens and smart contract validation.
- **Maintain privacy** through the use of ZKP, ensuring reviewers' anonymity.
- **Efficiently calculate reputation scores** using weighted algorithms embedded in Fabric chaincode.

Compared to traditional centralized reputation systems, the blockchain-based approach showed significant improvement in terms of **trustworthiness, resilience to manipulation, and auditability**. The feedback submission process remained fast and scalable, thanks to the modular design of the permissioned blockchain infrastructure.

V. CONCLUSION

This paper presented **Trust Chain**, a blockchain-powered reputation system for e-commerce platforms, addressing the key limitations of conventional feedback models. By combining **Hyperledger Indy** for decentralized identity management and **Hyperledger Fabric** for tamper-resistant feedback storage and smart contract execution, the system achieved **privacy preservation**, **security**, and **transparency**.

The inclusion of **feedback and discount tokens** ensured user accountability and incentivized honest participation. Additionally, the implementation of **Zero-Knowledge Proofs** allowed users to validate their credentials without compromising anonymity. This approach provides a scalable and efficient foundation for next-generation e-commerce platforms to build reliable trust frameworks, laying the groundwork for broader adoption in decentralized marketplaces.

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	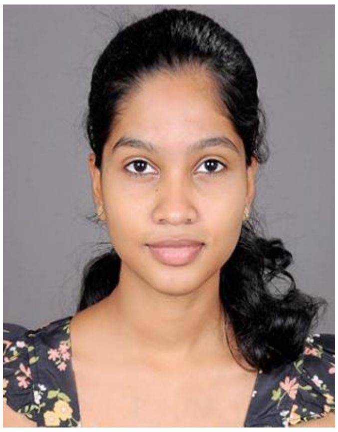		
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